

EGR 3350-07

Technical Communications for Engineers and Computer Scientists

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Process Description

Two-Factor Authentication Verification Process

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Introduction

Two-factor authentication is a security verification process that confirms a user's identity by requiring two separate forms of authentication credentials before granting access to a system or account. The process needs two different types of authentication which include both a password that the user knows and a mobile device or authentication application that the user owns. The purpose of two-factor authentication exists to enhance account protection through implementation of a second verification process which decreases the chances of unauthorized entry. The process occurs in five sequential steps.

Discussion

Figure 1 illustrates the sequence of steps in the two-factor authentication process.

Entry of Primary Credentials

The process begins when a user attempts to access a secure system by entering a username and password. These credentials are submitted through a login interface. The system compares the entered information with stored data in its authentication database. If the credentials are valid, the process advances to the second stage. If the credentials are incorrect, access is denied and the authentication attempt ends.

System Validation and Code Generation

The system produces a temporary verification code after the main credentials have been checked. The code is created randomly and remains active for a period that lasts between 30 seconds and 60 seconds. The code exists as a temporary security measure which stops any attempt to reuse captured or used security codes. The process provides a second security measure which operates in addition to existing password protection.

Transmission of Verification Code

The system sends the generated code to the user through an alternative authentication method.

The system supports three common transmission methods which include sending text messages and emails and using authentication applications that run on mobile devices. The system improves account security by sending the second authentication factor to a device that the authorized user currently owns.

Entry and Verification of Secondary Code

The user starts the authentication process by entering the verification code which he received.

The system compares the submitted code with the originally generated value. The authentication process continues when the code matches, and the code has not expired. The system denies access when the code either contains invalid information, or it has reached its expiration date which requires the user to restart the login process.

Access Authorization

The system gives account access to users after both authentication factors have been successfully verified. The user can continue to use the system after this point. The system denies access when either authentication factor fails to pass verification. The two-step authentication process requires users to provide both a password and their physical presence to gain access.

Conclusion

Two-factor authentication is a security verification process that confirms a user's identity by requiring two separate forms of authentication credentials before granting access to a system or account. The process starts with credential validation and moves to temporary code generation and verification before ending with access authorization. Two-factor authentication uses

knowledge-based and possession-based authentication methods to decrease unauthorized access risk while allowing legitimate users to maintain system access.

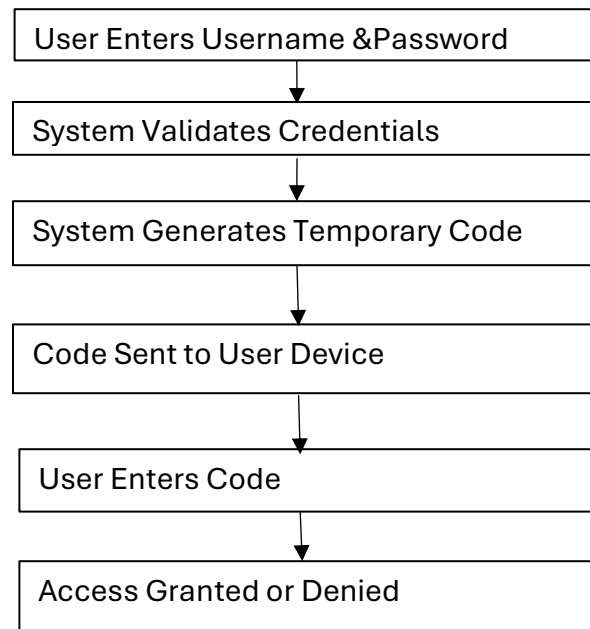


Figure 1. Two-Factor Authentication verification process. Source: Author.